



**ENGINEERED
TO PROTECT.
BUILT TO PERFORM.**

**High-Precision Thermal
Packaging & Phase Change
Materials for Critical Cold Chain**



ABOUT US

Alpine Thermal Solutions is a highly specialized, engineering-driven manufacturer of passive cooling solutions. Located strategically within the Hosur-Bengaluru industrial hub, we formulate, extrude, and manufacture advanced thermal barriers designed to safeguard high-value pharmaceuticals, biologic, clinical trial samples, and premium frozen food exports.



CORE PRODUCT PORTFOLIO

1. Advanced Temperature-Controlled Gel Packs

Engineered for high-volume, reliable 0°C transit environments.

- Leak-Proof Architecture
- Precision Integrity
- Flexible Configurations

2. Phase Change Material (PCM) Thermal Panels

Rigid, high-density thermal storage units designed for multi-use, circular supply chains.

- Structural Innovation
- Heavy-Duty Construction
- Leak-Free Sealing

3. Tailored Eutectic Formulations

We eliminate the risks of "over-chilling" or premature thawing by chemically locking exact thermal plateaus:

- **The Chilled Range (+5°C):** Salt-hydrate chemistry optimized with advanced nucleating agents to eliminate supercooling, keeping vaccines precisely in the +2°C to +8°C safety window.
- **The Deep-Freeze Range (-18°C to -24°C):** High-latent-heat Magnesium Chloride eutectic configurations designed as sustainable, reusable alternatives to hazardous dry ice.

THE ALPINE EDGE

Custom Polymer & Chemical Integration: Because we control both the multi-layer film extrusion and the chemical gel formulation, we deliver perfectly harmonized packaging systems that outperform commodity ice packs.

Salt-Tolerant Gel Stability: Our proprietary thickeners guarantee that heavy salt-hydrate solutions remain perfectly in suspension over hundreds of freeze-thaw cycles without settling or performance decay.

Rigorous Thermal Profiling: We back our products with data-logged temperature-over-time validation graphs, proving cold chain compliance before your cargo leaves the dock.

THE ALPINE EDGE

Optimize your thermal packaging configurations today: Contact our engineering team to request sample kits or to coordinate a data-logged thermal profiling test tailored to your exact shipping shippers and payloads.

Contact us:

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